

lead_quality_analysis

June 16, 2026

```
[47]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import chi2_contingency, fisher_exact
import statsmodels.api as sm
import statsmodels.formula.api as smf
```

1 Data Loading and Initial Exploration

The first step is to load the lead dataset and inspect its structure. This allows us to verify the available columns, identify potential data quality issues, and understand how lead outcomes are recorded.

Key objectives: - Load the dataset into a pandas DataFrame - Review column names and data types - Confirm row counts and overall dataset size - Identify fields relevant to lead quality analysis

```
[48]: # Load data
file_path = "Data/Analyst_case_study_dataset_1_(1).xls"

df = pd.read_excel(file_path)

df.head()
```

```
[48]:      LeadCreated  FirstName      Email \
0 2009-07-01 01:24:29.563    Dorinda  kanani@sandwichisles.net
1 2009-04-13 15:27:35.990   Presetta  clerk2@ustconline.net
2 2009-04-21 07:22:59.000      Gina  wagoner_gina@yahoo.com
3 2009-08-03 19:39:02.000      Kari   usa4ley@yahoo.com
4 2009-04-13 10:18:37.490  Stephanie  sr1lambert@embarqmail.com
```

```
      VendorLeadID \
0  FDF81FDA-A649-437B-B99C-FDDE74F7971B
1  4190ACB7-5026-416C-B987-ED8AD427D5E6
2           hFg80jf_R0CRN55hdhWILw
3           jB01QgYZxkWArI9jWxuufw
4  D5B32074-458E-40EC-B185-1FEF20AC626D
```

```
      CallStatus \
```

0	NaN
1	NaN
2	Unable to contact - Bad Contact Information
3	Contacted - Doesn't Qualify
4	NaN

	WidgetName	PublisherZoneName	\
0	w-302252-DebtReduction1-1DC-CreditSolutions	TopLeft-302252	
1	w-300250-DebtReduction1-1DC-Head2	TopLeft-302252	
2	w-300250-DebtReduction1-1DC-Head2	TopLeft-302252	
3	w-302252-DebtReduction1-1DC-white	TopLeft-302252	
4	w-300250-DebtReduction1-1DC-BlueMeter	TopLeft-302252	

	PublisherCampaignName	AddressScore	PhoneScore	...	Partner	\
0	DebtReductionInc	NaN	5.0	...	google	
1	DebtReductionInc	NaN	NaN	...	AdKnowledge	
2	DebtReductionInc	NaN	NaN	...	AdKnowledge	
3	DebtReductionInc	5.0	3.0	...	Google	
4	DebtReductionInc	NaN	NaN	...	Google	

	ReferralDomain	MarketingCampaign	AdGroup	\
0	www.google.com	Debt Holding Tank	Holding Tank - Debt	
1	NaN	Financial Services	Consolidate	
2	us.mc582.mail.yahoo.com	Financial Services	Consolidate	
3	norwich.kijiji.com	DebtReductionInc	Lower Payments	
4	NaN	DebtReductionInc	Debt Reduction	

	Keyword	SearchQuery	\
0	Debt specialists	debt specialists	
1	NaN	NaN	
2	NaN	NaN	
3	NaN	NaN	
4	NaN	NaN	

	ReferralURL	\
0	http://www.google.com/search	
1	NaN	
2	http://us.mc582.mail.yahoo.com/mc/showMessage	
3	http://norwich.kijiji.com/c-Cars-vehicles-Cars...	
4	NaN	

	ReferralURL Parameters	\
0	sourceid=navclient&aq=1&oq=debt sp&ie=UTF-8&rl...	
1	NaN	
2	&fid=Inbox&sort=date&order=down&startMid=0&.ra...	
3	NaN	
4	NaN	

```

                                LandingPageURL  \
0  http://www.debtredutioninc.com/index8.html
1  http://www.debtredutioninc.com/index8.html
2  http://www.debtredutioninc.com/index8.html
3  http://www.debtredutioninc.com/index12.html
4  http://www.debtredutioninc.com/index8.html

```

```

                                Landing Page URL Parameters
0  utm_source=google&utm_medium=CPC&utm_content=H...
1  utm_source=AdKnowledge&utm_medium=CPC&utm_cont...
2  utm_source=AdKnowledge&utm_medium=CPC&utm_cont...
3  utm_source=Google&utm_medium=cpc&utm_campaign=...
4  utm_source=Google&utm_medium=cpc&utm_campaign=...

```

[5 rows x 24 columns]

```

[49]: # Clean column names
df.columns = (
    df.columns
    .str.strip()
    .str.replace(" ", "_")
    .str.replace("-", "_")
)

df.columns

```

```

[49]: Index(['LeadCreated', 'FirstName', 'Email', 'VendorLeadID', 'CallStatus',
            'WidgetName', 'PublisherZoneName', 'PublisherCampaignName',
            'AddressScore', 'PhoneScore', 'AdvertiserCampaignName', 'State',
            'DebtLevel', 'IP_Address', 'Partner', 'ReferralDomain',
            'MarketingCampaign', 'AdGroup', 'Keyword', 'SearchQuery', 'ReferralURL',
            'ReferralURL_Parameters', 'LandingPageURL',
            'Landing_Page_URL_Parameters'],
            dtype='object')

```

2 Data Cleaning and Feature Engineering

Before analysis, the dataset requires standardization of column names and creation of lead quality indicators.

Based on the advertiser's definitions:

Good Lead Outcomes: - Closed - EP Sent - EP Received - EP Confirmed

Bad Lead Outcomes: - Unable to Contact - Contacted - Invalid Profile - Contacted - Doesn't Qualify

Unknown Outcomes: - All remaining statuses

Binary indicators are created to support statistical analysis and segmentation.

```
[50]: df["CallStatus_clean"] = df["CallStatus"].fillna("Unknown").str.strip()

good_status = [
    "Closed",
    "EP Sent",
    "EP Received",
    "EP Confirmed"
]

bad_status = [
    "Unable to contact - Bad Contact Information",
    "Contacted - Invalid Profile",
    "Contacted - Doesn't Qualify"
]

df["is_good"] = df["CallStatus_clean"].isin(good_status).astype(int)
df["is_closed"] = (df["CallStatus_clean"] == "Closed").astype(int)
df["is_bad"] = df["CallStatus_clean"].isin(bad_status).astype(int)
df["is_unknown"] = (~df["is_good"].astype(bool) & ~df["is_bad"].astype(bool)).
    ↳astype(int)
```

3 Overall Lead Quality Performance

This section establishes baseline performance across the entire dataset.

Metrics: - Total Leads - Good Leads - Closed Leads - Bad Leads - Quality Rate - Closed Rate

These metrics serve as benchmarks when evaluating individual segments later in the analysis.

```
[51]: # Overall lead quality
overall = pd.DataFrame({
    "total_leads": [len(df)],
    "good_leads": [df["is_good"].sum()],
    "closed_leads": [df["is_closed"].sum()],
    "bad_leads": [df["is_bad"].sum()],
    "unknown_leads": [df["is_unknown"].sum()],
    "quality_rate": [df["is_good"].mean()],
    "closed_rate": [df["is_closed"].mean()]
})

overall
```

```
[51]:   total_leads  good_leads  closed_leads  bad_leads  unknown_leads  \
0          3021          393           245          488           2140

   quality_rate  closed_rate
```

0 0.130089 0.081099

4 Lead Quality Trend Analysis

To determine whether lead quality is improving or declining over time, leads are aggregated by week.

Metrics tracked: - Weekly Lead Volume - Weekly Quality Rate - Weekly Closed Rate

The advertiser is primarily interested in lead quality improvements, so trend analysis helps identify whether performance is moving toward or away from business objectives.

```
[52]: # Trend over time by week
df["week"] = df["LeadCreated"].dt.to_period("W").dt.start_time

weekly = (
    df.groupby("week")
    .agg(
        leads=("VendorLeadID", "count"),
        good_leads=("is_good", "sum"),
        closed_leads=("is_closed", "sum"),
        bad_leads=("is_bad", "sum")
    )
    .reset_index()
)

weekly["quality_rate"] = weekly["good_leads"] / weekly["leads"]
weekly["closed_rate"] = weekly["closed_leads"] / weekly["leads"]

weekly
```

```
[52]:
```

	week	leads	good_leads	closed_leads	bad_leads	quality_rate	\
0	2009-03-30	37	6	3	3	0.162162	
1	2009-04-06	119	18	15	9	0.151261	
2	2009-04-13	125	21	16	10	0.168000	
3	2009-04-20	154	25	14	22	0.162338	
4	2009-04-27	129	14	10	14	0.108527	
5	2009-05-04	113	13	10	10	0.115044	
6	2009-05-11	165	21	12	20	0.127273	
7	2009-05-18	122	11	8	11	0.090164	
8	2009-05-25	132	9	5	15	0.068182	
9	2009-06-01	137	14	10	30	0.102190	
10	2009-06-08	164	25	17	37	0.152439	
11	2009-06-15	151	25	21	40	0.165563	
12	2009-06-22	107	12	9	25	0.112150	
13	2009-06-29	100	17	9	17	0.170000	
14	2009-07-06	104	10	8	23	0.096154	
15	2009-07-13	115	10	6	30	0.086957	

16	2009-07-20	144	13	11	28	0.090278
17	2009-07-27	136	13	9	22	0.095588
18	2009-08-03	95	19	11	16	0.200000
19	2009-08-10	116	20	14	17	0.172414
20	2009-08-17	103	10	4	18	0.097087
21	2009-08-24	88	8	7	13	0.090909
22	2009-08-31	74	10	3	10	0.135135
23	2009-09-07	65	16	5	12	0.246154
24	2009-09-14	95	12	4	18	0.126316
25	2009-09-21	82	15	4	8	0.182927
26	2009-09-28	41	6	0	10	0.146341

	closed_rate
0	0.081081
1	0.126050
2	0.128000
3	0.090909
4	0.077519
5	0.088496
6	0.072727
7	0.065574
8	0.037879
9	0.072993
10	0.103659
11	0.139073
12	0.084112
13	0.090000
14	0.076923
15	0.052174
16	0.076389
17	0.066176
18	0.115789
19	0.120690
20	0.038835
21	0.079545
22	0.040541
23	0.076923
24	0.042105
25	0.048780
26	0.000000

4.1 Weekly Lead Quality Trend

This chart shows how lead quality and closed rates changed over time.

The dashed horizontal line represents the advertiser's target quality rate of 9.6%.

This helps determine whether lead quality is moving toward the advertiser's proposed CPL increase target.

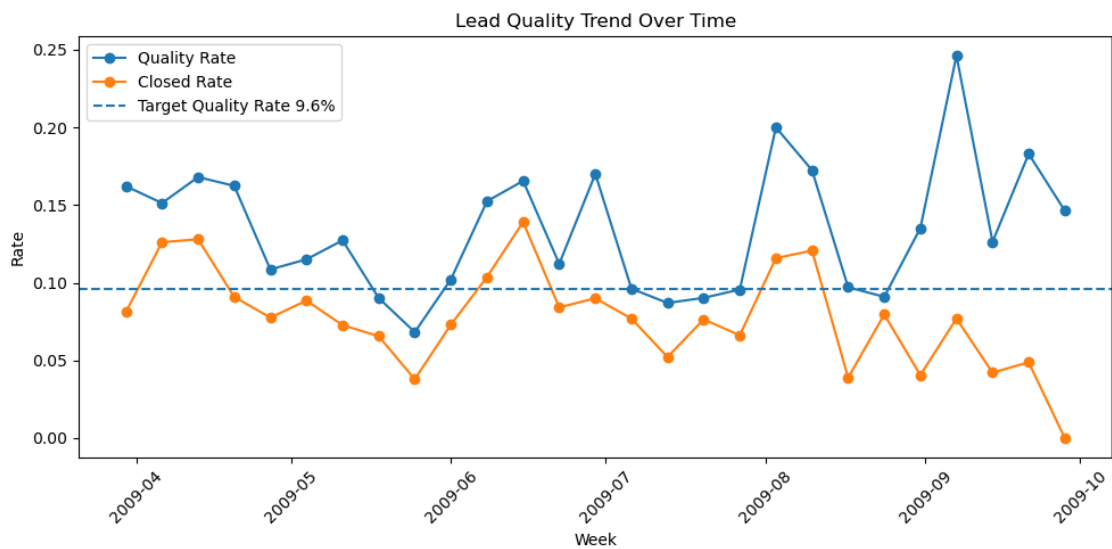
```
[53]: plt.figure(figsize=(10,5))

x = weekly["week"].to_numpy()
quality = weekly["quality_rate"].to_numpy()
closed = weekly["closed_rate"].to_numpy()

plt.plot(x, quality, marker="o", label="Quality Rate")
plt.plot(x, closed, marker="o", label="Closed Rate")

plt.axhline(y=0.096, linestyle="--", label="Target Quality Rate 9.6%")

plt.title("Lead Quality Trend Over Time")
plt.xlabel("Week")
plt.ylabel("Rate")
plt.legend()
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



Question 1: Is Lead Quality Improving Over Time?

4.2 Statistical Test for Time Trends

Visual trends can sometimes be misleading. To determine whether lead quality is truly improving or declining, a logistic regression model is fitted using lead quality as the outcome variable and lead creation date as the predictor.

Interpretation: - Positive coefficient = improving quality - Negative coefficient = declining quality
 - p-value < 0.05 = statistically significant

```
[54]: trend_df = df.dropna(subset=["LeadCreated"]).copy()

trend_df["date_num"] = (
    trend_df["LeadCreated"]
    - trend_df["LeadCreated"].min()
).dt.days

model = smf.logit(
    "is_good ~ date_num",
    data=trend_df
).fit()

model.summary()
```

Optimization terminated successfully.
 Current function value: 0.386545
 Iterations 6

```
[54]:
```

Dep. Variable:	is_good	No. Observations:	3021
Model:	Logit	Df Residuals:	3019
Method:	MLE	Df Model:	1
Date:	Tue, 16 Jun 2026	Pseudo R-squ.:	3.030e-05
Time:	17:44:02	Log-Likelihood:	-1167.8
converged:	True	LL-Null:	-1167.8
Covariance Type:	nonrobust	LLR p-value:	0.7902

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-1.9242	0.105	-18.263	0.000	-2.131	-1.718
date_num	0.0003	0.001	0.266	0.790	-0.002	0.002

```
[55]: coef = model.params["date_num"]
pval = model.pvalues["date_num"]

print("Coefficient:", round(coef,6))
print("P-value:", round(pval,6))

if pval < 0.05:
    if coef > 0:
        print("Lead quality is improving over time.")
    else:
        print("Lead quality is declining over time.")
else:
    print("No statistically significant trend detected.")
```

Coefficient: 0.000289
 P-value: 0.79015
 No statistically significant trend detected.

Question 2: What Drives Lead Quality?

4.3 Segment Analysis

Lead quality may vary across different traffic sources, creative designs, customer characteristics, and campaigns.

The analysis below evaluates quality rates across key dimensions and identifies segments that outperform the overall average.

```
[56]: def segment_quality(df, col, min_leads=30):

    result = (
        df.groupby(col)
        .agg(
            leads=("VendorLeadID", "count"),
            good_leads=("is_good", "sum"),
            closed_leads=("is_closed", "sum")
        )
        .reset_index()
    )

    result["quality_rate"] = (
        result["good_leads"] / result["leads"]
    )

    result["closed_rate"] = (
        result["closed_leads"] / result["leads"]
    )

    overall_quality = df["is_good"].mean()

    result["lift_vs_avg"] = (
        result["quality_rate"] / overall_quality - 1
    )

    return (
        result[result["leads"] >= min_leads]
        .sort_values("quality_rate", ascending=False)
    )
```

4.3.1 Widget Performance

WidgetName represents different creative designs and layouts shown to users.

If certain creatives consistently generate higher quality leads, traffic can be shifted toward those designs.

```
[57]: widget_results = segment_quality(
    df,
    "WidgetName"
```

```
)

widget_results.head(20)
```

```
[57]:
```

	WidgetName	leads	good_leads	\
11	w-302252-DebtReduction1-1DC-yellowarrow	49	12	
2	w-300250-DebtReduction1-1DC-CreditSolutions	76	18	
0	w-300250-DebtReduction1-1DC	348	56	
3	w-300250-DebtReduction1-1DC-Head2	89	14	
1	w-300250-DebtReduction1-1DC-BlueMeter	92	14	
6	w-300250-DebtReduction1-2DC-BlueMeter	87	12	
12	w-302252-DebtReduction1-1DC-yellowarrow-blue	230	31	
9	w-302252-DebtReduction1-1DC-CreditSolutions	1052	131	
8	w-302252-DebtReduction1-1DC	271	32	
13	w-302252-DebtReduction1-1DC-yellowarrow-dark	135	15	
7	w-300250-DebtReduction1-2DC-CreditSolutions	75	8	
10	w-302252-DebtReduction1-1DC-white	428	44	
4	w-300250-DebtReduction1-1DC-Head3	75	5	

	closed_leads	quality_rate	closed_rate	lift_vs_avg
11	3	0.244898	0.061224	0.882536
2	12	0.236842	0.157895	0.820611
0	34	0.160920	0.097701	0.236992
3	11	0.157303	0.123596	0.209195
1	13	0.152174	0.141304	0.169764
6	6	0.137931	0.068966	0.060279
12	15	0.134783	0.065217	0.036077
9	76	0.124525	0.072243	-0.042776
8	23	0.118081	0.084871	-0.092307
13	10	0.111111	0.074074	-0.145886
7	6	0.106667	0.080000	-0.180051
10	31	0.102804	0.072430	-0.209745
4	4	0.066667	0.053333	-0.487532

4.3.2 Campaign Performance

Campaign-level analysis helps identify which acquisition channels generate the highest quality leads.

```
[58]: campaign_results = segment_quality(
    df,
    "MarketingCampaign"
)

campaign_results.head(20)
```

```
[58]:
```

	MarketingCampaign	leads	good_leads	closed_leads	quality_rate	\
17	Financial Services	171	32	21	0.187135	
5	Debt Holding Tank	337	61	49	0.181009	

7	Debt Volume	143	23	13	0.160839
0	Credit	100	14	12	0.140000
18	state	97	12	10	0.123711
3	Debt Consolidation	34	4	1	0.117647
11	DebtReductionInc	1468	166	88	0.113079
4	Debt General	328	29	20	0.088415

	closed_rate	lift_vs_avg
17	0.122807	0.438507
5	0.145401	0.391420
7	0.090909	0.236374
0	0.120000	0.076183
18	0.103093	-0.049028
3	0.029412	-0.095644
11	0.059946	-0.130759
4	0.060976	-0.320355

4.3.3 Geographic Performance

Lead quality may vary by state due to differences in consumer behavior, qualification requirements, or debt characteristics.

```
[59]: state_results = segment_quality(
      df,
      "State"
    )

state_results.head(20)
```

```
[59]:   State  leads  good_leads  closed_leads  quality_rate  closed_rate  \
24   OK     50         13           9         0.260000         0.180000
9    HI     32          7           5         0.218750         0.156250
14   MA     70         12           4         0.171429         0.057143
1    AL     95         16          12         0.168421         0.126316
4    CA    462        73          44         0.158009         0.095238
31   WV     52          8           5         0.153846         0.096154
25   OR     53          8           3         0.150943         0.056604
3    AZ     94         14           9         0.148936         0.095745
22   NV     54          8           5         0.148148         0.092593
29   VA    145        21          12         0.144828         0.082759
28   TX    323        42          28         0.130031         0.086687
8    FL    261        33          18         0.126437         0.068966
16   MI    145        17          11         0.117241         0.075862
5    CO    105        12           7         0.114286         0.066667
26   PA     75          8           6         0.106667         0.080000
2    AR     48          5           3         0.104167         0.062500
17   MO     89          9           6         0.101124         0.067416
13   LA     80          8           5         0.100000         0.062500
```

23	NY	283	28	17	0.098940	0.060071
11	IL	184	18	14	0.097826	0.076087

	lift_vs_avg
24	0.998626
9	0.681536
14	0.317775
1	0.294656
4	0.214616
31	0.182619
25	0.160305
3	0.144876
22	0.138818
29	0.113293
28	-0.000449
8	-0.028078
16	-0.098763
5	-0.121483
26	-0.180051
2	-0.199268
17	-0.222661
13	-0.231298
23	-0.239446
11	-0.248009

4.3.4 Address Verification Quality

AddressScore measures how closely the submitted address matches external records.

Higher scores may indicate more trustworthy lead information.

```
[60]: address_results = segment_quality(
      df,
      "AddressScore",
      min_leads=20
    )

address_results
```

[60]:	AddressScore	leads	good_leads	closed_leads	quality_rate	closed_rate	\
1	2.0	38	7	4	0.184211	0.105263	
4	5.0	868	120	64	0.138249	0.073733	
0	1.0	122	14	5	0.114754	0.040984	
2	3.0	93	9	5	0.096774	0.053763	
3	4.0	48	2	1	0.041667	0.020833	

	lift_vs_avg
1	0.416031

```

4      0.062722
0     -0.117883
2     -0.256095
3     -0.679707

```

```

[61]: phone_results = segment_quality(
      df,
      "PhoneScore",
      min_leads=20
    )

phone_results

```

```

[61]:   PhoneScore  leads  good_leads  closed_leads  quality_rate  closed_rate  \
4         5.0    619         100          54      0.161551      0.087237
2         3.0    426          50          28      0.117371      0.065728
1         2.0    160          16           7      0.100000      0.043750
3         4.0    184          13           7      0.070652      0.038043

```

```

      lift_vs_avg
4      0.241845
2     -0.097767
1     -0.231298
3     -0.456895

```

4.3.5 Debt Level Analysis

Consumers with higher debt levels may be more likely to qualify for debt relief programs.

Debt levels are grouped into ranges for comparison.

```

[62]: df["DebtLevel"].value_counts(dropna=False).head(20)

debt_results = segment_quality(
    df,
    "DebtLevel",
    min_leads=20
)

debt_results

```

```

[62]:   DebtLevel  leads  good_leads  closed_leads  quality_rate  \
5    70001-90000    131         25          18      0.190840
0    10001-15000    291         54          34      0.185567
2    20001-30000    454         72          40      0.158590
4    50001-70000    244         38          22      0.155738
1    15001-20000    408         57          35      0.139706
3    30001-50000    495         61          39      0.123232

```

8	90000-100000	90	11	9	0.122222
7	7500-15000	270	31	19	0.114815
6	7500-10000	440	31	22	0.070455
9	More_than_100000	190	13	7	0.068421

	closed_rate	lift_vs_avg
5	0.137405	0.466989
0	0.116838	0.426458
2	0.088106	0.219087
4	0.090164	0.197159
1	0.085784	0.073922
3	0.078788	-0.052710
8	0.100000	-0.060475
7	0.070370	-0.117416
6	0.050000	-0.458414
9	0.036842	-0.474046

Question 3: Can We Reach 9.6% Quality?

4.4 Opportunity Analysis

The advertiser is willing to increase CPL by 20% if lead quality improves from 8.0% to 9.6%.

The objective is to identify segments already exceeding the target quality rate and determine whether traffic can be reallocated toward those segments.

```
[63]: target_rate = 0.096

opportunities = []

for column in [
    "WidgetName",
    "MarketingCampaign",
    "State",
    "AddressScore",
    "PhoneScore",
    "debt_bucket"
]:

    if column in df.columns:

        temp = segment_quality(
            df,
            column,
            min_leads=30
        )

        temp = temp[
```

```

        temp["quality_rate"] >= target_rate
    ]

    temp["segment"] = column

    opportunities.append(temp)

opportunity_df = pd.concat(
    opportunities,
    ignore_index=True
)

opportunity_df.sort_values(
    "quality_rate",
    ascending=False
).head(30)

```

```

[63]:

```

	WidgetName	leads	good_leads	\
19	NaN	50	13	
0	w-302252-DebtReduction1-1DC-yellowarrow	49	12	
1	w-300250-DebtReduction1-1DC-CreditSolutions	76	18	
20	NaN	32	7	
12	NaN	171	32	
39	NaN	38	7	
13	NaN	337	61	
21	NaN	70	12	
22	NaN	95	16	
43	NaN	619	100	
2	w-300250-DebtReduction1-1DC	348	56	
14	NaN	143	23	
23	NaN	462	73	
3	w-300250-DebtReduction1-1DC-Head2	89	14	
24	NaN	52	8	
4	w-300250-DebtReduction1-1DC-BlueMeter	92	14	
25	NaN	53	8	
26	NaN	94	14	
27	NaN	54	8	
28	NaN	145	21	
15	NaN	100	14	
40	NaN	868	120	
5	w-300250-DebtReduction1-2DC-BlueMeter	87	12	
6	w-302252-DebtReduction1-1DC-yellowarrow-blue	230	31	
29	NaN	323	42	
30	NaN	261	33	
7	w-302252-DebtReduction1-1DC-CreditSolutions	1052	131	
16	NaN	97	12	
8	w-302252-DebtReduction1-1DC	271	32	

17

NaN

34

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	closed_leads	quality_rate	closed_rate	lift_vs_avg	segment \
19	9	0.260000	0.180000	0.998626	State
0	3	0.244898	0.061224	0.882536	WidgetName
1	12	0.236842	0.157895	0.820611	WidgetName
20	5	0.218750	0.156250	0.681536	State
12	21	0.187135	0.122807	0.438507	MarketingCampaign
39	4	0.184211	0.105263	0.416031	AddressScore
13	49	0.181009	0.145401	0.391420	MarketingCampaign
21	4	0.171429	0.057143	0.317775	State
22	12	0.168421	0.126316	0.294656	State
43	54	0.161551	0.087237	0.241845	PhoneScore
2	34	0.160920	0.097701	0.236992	WidgetName
14	13	0.160839	0.090909	0.236374	MarketingCampaign
23	44	0.158009	0.095238	0.214616	State
3	11	0.157303	0.123596	0.209195	WidgetName
24	5	0.153846	0.096154	0.182619	State
4	13	0.152174	0.141304	0.169764	WidgetName
25	3	0.150943	0.056604	0.160305	State
26	9	0.148936	0.095745	0.144876	State
27	5	0.148148	0.092593	0.138818	State
28	12	0.144828	0.082759	0.113293	State
15	12	0.140000	0.120000	0.076183	MarketingCampaign
40	64	0.138249	0.073733	0.062722	AddressScore
5	6	0.137931	0.068966	0.060279	WidgetName
6	15	0.134783	0.065217	0.036077	WidgetName
29	28	0.130031	0.086687	-0.000449	State
30	18	0.126437	0.068966	-0.028078	State
7	76	0.124525	0.072243	-0.042776	WidgetName
16	10	0.123711	0.103093	-0.049028	MarketingCampaign
8	23	0.118081	0.084871	-0.092307	WidgetName
17	1	0.117647	0.029412	-0.095644	MarketingCampaign

	MarketingCampaign	State	AddressScore	PhoneScore
19		NaN	OK	NaN
0		NaN	NaN	NaN
1		NaN	NaN	NaN
20		NaN	HI	NaN
12	Financial Services	NaN	NaN	NaN
39		NaN	NaN	2.0
13	Debt Holding Tank	NaN	NaN	NaN
21		NaN	MA	NaN
22		NaN	AL	NaN
43		NaN	NaN	5.0
2		NaN	NaN	NaN
14	Debt Volume	NaN	NaN	NaN

23		NaN	CA	NaN	NaN
3		NaN	NaN	NaN	NaN
24		NaN	WV	NaN	NaN
4		NaN	NaN	NaN	NaN
25		NaN	OR	NaN	NaN
26		NaN	AZ	NaN	NaN
27		NaN	NV	NaN	NaN
28		NaN	VA	NaN	NaN
15	Credit	NaN	NaN	NaN	NaN
40		NaN	NaN	5.0	NaN
5		NaN	NaN	NaN	NaN
6		NaN	NaN	NaN	NaN
29		NaN	TX	NaN	NaN
30		NaN	FL	NaN	NaN
7		NaN	NaN	NaN	NaN
16	state	NaN	NaN	NaN	NaN
8		NaN	NaN	NaN	NaN
17	Debt Consolidation	NaN	NaN	NaN	NaN